THE LIMIT OF AFFINES WITH AFFINE TRANSITIONS IS AFFINE (QUASI-AFFINE ROUTE)

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Limit closed nonempty. Limits, Lemma lemma-limit-closed-nonempty
- Topology of limits. Limits, topology lemmas of the chapter

Source reference. Original reference 086P: sources/stacks-chow/tags.txt, source lines 10399–10399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z5: sources/stacks-chow/tags.txt, source lines 2580–2580. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z0: sources/stacks-chow/tags.txt, source lines 2575–2575. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01Z3: sources/stacks-chow/tags.txt, source lines 2578–2578. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YY: sources/stacks-chow/tags.txt, source lines 2573–2573. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 086P if S is affine, then for some $i_0 \in I$ the schemes S_i for $i \geq i_0$ are affine.*

Proof. By Lemma 01Z5 we may assume that S_0 is quasi-affine for some $0 \in I$. Set $R_0 = \Gamma(S_0, \mathcal{O}_{S_0})$. Then S_0 is a quasi-compact open of $T_0 = \mathrm{Spec}(R_0)$. Denote $j_0 : S_0 \to T_0$ the corresponding quasi-compact open immersion. For $i \geq 0$ set $\mathcal{A}_i = f_{i0,*}\mathcal{O}_{S_i}$. Since f_{i0} is affine we see that $S_i = \underline{\mathrm{Spec}}_{S_0}(\mathcal{A}_i)$. Set $T_i = \underline{\mathrm{Spec}}_{T_0}(j_{0,*}\mathcal{A}_i)$. Then $T_i \to T_0$ is affine, hence T_i is affine. Thus T_i is the spectrum of

$$R_i = \Gamma(T_0, j_{0,*}\mathcal{A}_i) = \Gamma(S_0, \mathcal{A}_i) = \Gamma(S_i, \mathcal{O}_{S_i}).$$

Write $S = \mathrm{Spec}(R)$. We have $R = \mathrm{colim}_i R_i$ by Lemma 01Z0. Hence also $S = \lim_i T_i$. As formation of the relative spectrum commutes with base change, the inverse image of the open $S_0 \subset T_0$ in T_i is S_i. Let $Z_0 = T_0 \setminus S_0$ and let $Z_i \subset T_i$ be the inverse image of Z_0. As $S_i = T_i \setminus Z_i$, it suffices to show that Z_i is empty for some i. Assume Z_i is nonempty for all i to get a contradiction. By Lemma 01Z3 there exists a point s of $S = \lim T_i$ which maps to a point of Z_i for every i. But $S = \lim_i S_i$, and hence we arrive at a contradiction by Lemma 01YY. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.